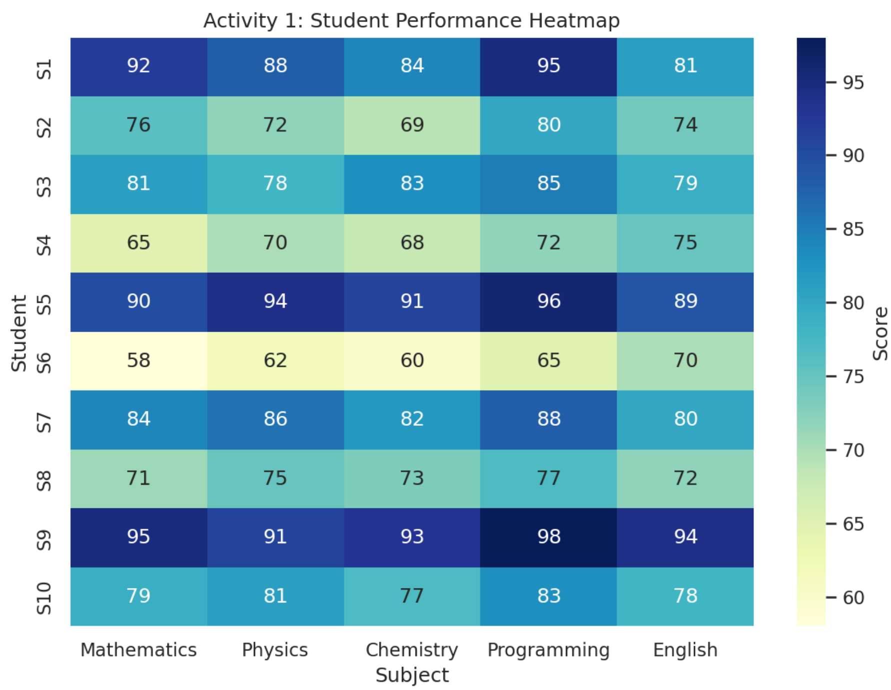
Visualization Activities with Datasets — Solutions

Data Handling, Visualization & Graphical Analysis

Activity 1: Heatmap – Student Performance Analysis

A heatmap uses color intensity to represent the magnitude of each score across students (rows) and subjects (columns), making patterns and outliers easy to spot at a glance.



Q1. Construct a Heatmap.

The heatmap above plots all 10 students against the 5 subjects, with darker/lighter shading (YlGnBu color scale) representing lower/higher marks respectively, and exact scores annotated in each cell.

Q2. Identify highest overall performer.

Student S9 has the highest overall average at 94.2 (Math 95, Physics 91, Chemistry 93, Programming 98, English 94), followed closely by S5 (92.0).

Q3. Find highest average subject.

Programming has the highest average score across all students at 83.9, ahead of English (79.2), Physics (79.7), Mathematics (79.1), and Chemistry (78.0). This suggests students found programming concepts relatively easier or were better prepared for it.

Q4. Identify students needing support.

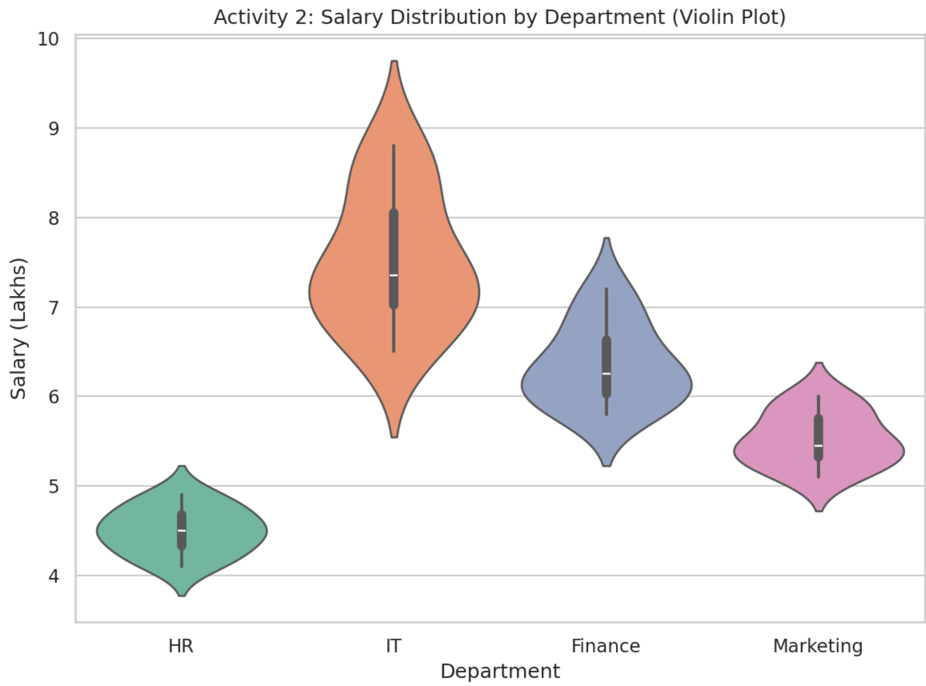
Based on overall averages, S6 (63.0), S4 (70.0), and S8 (73.6) are the weakest performers and need academic support. S6 in particular scores low across every subject (58–70 range), indicating a broad, not subject-specific, weakness.

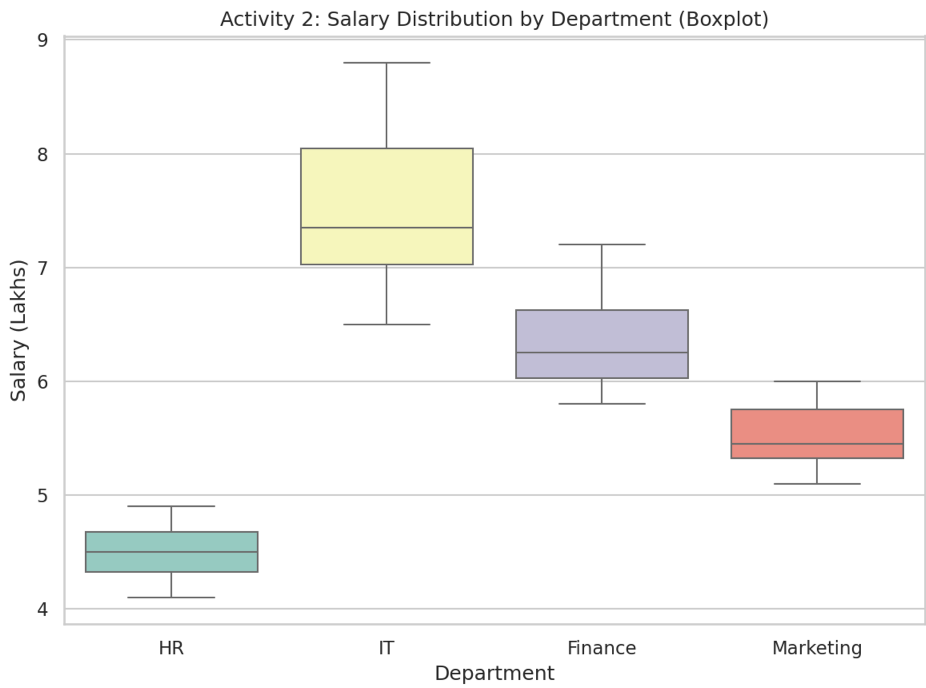
Q5. Explain Heatmap advantages.

Heatmaps compress a large numeric table into an intuitive visual: color gradients let a viewer instantly spot high/low performers and strong/weak subjects without reading every number. They scale well to large datasets, reveal rowwise and column-wise patterns simultaneously, and make outliers and clusters (e.g., a consistently weak student, or an easy subject) visually obvious in a way a raw table cannot.

Activity 2: Violin Plot & Boxplot – Employee Salary Analysis

Violin plots combine a boxplot with a rotated, mirrored kernel density plot, showing both summary statistics and the full shape of the salary distribution per department.





Q1. Draw violin plots.

The violin plot above shows salary distribution (in lakhs) for HR, IT, Finance, and Marketing. IT's violin is visibly the tallest/widest, reflecting its higher and more spread-out salaries.

Q2. Construct boxplots.

The boxplot above shows the median, interquartile range (box), and range (whiskers) of salaries for each department, allowing quick comparison of central tendency and spread.

Q3. Largest variation?

IT shows the largest variation, with a standard deviation of about 0.755 lakhs (range 6.5–8.8), compared to HR's tight 0.258, Finance's 0.455, and Marketing's 0.301.

Q4. Highest median?

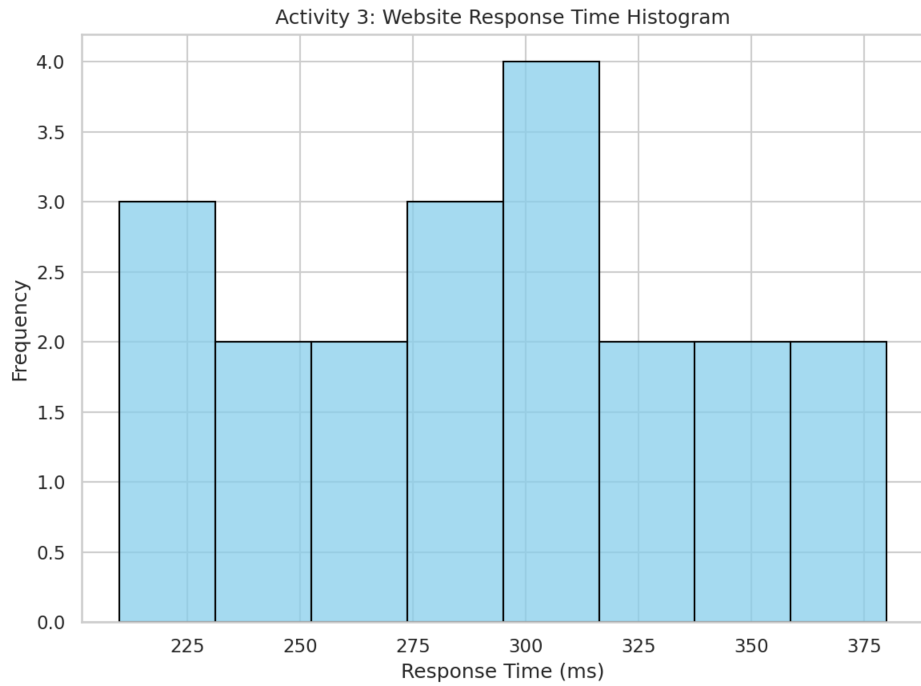
IT has the highest median salary at 7.35 lakhs, followed by Finance (6.25), Marketing (5.45), and HR (4.50).

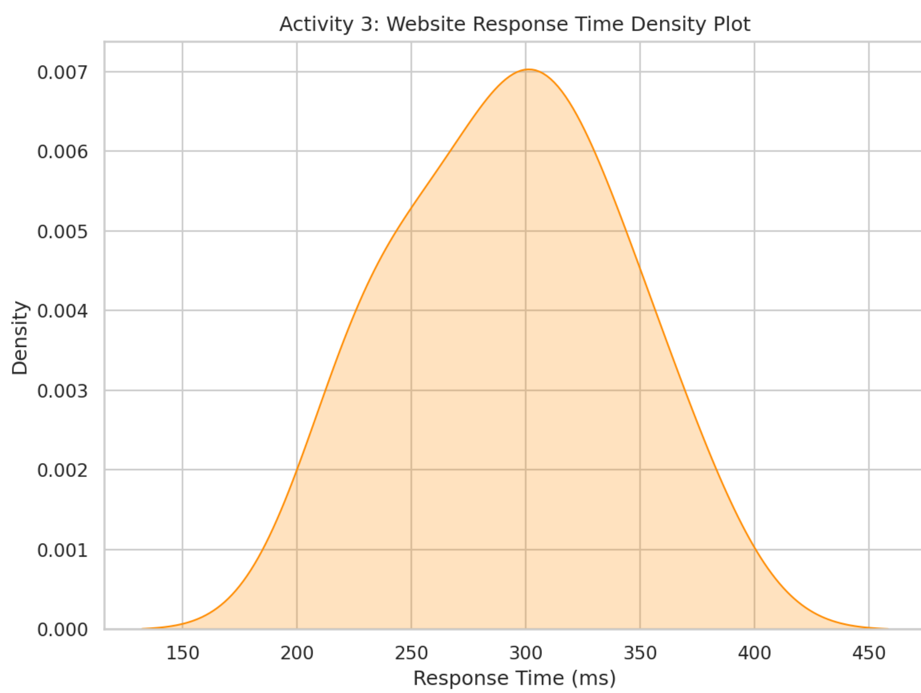
Q5. Which plot better represents distribution?

The violin plot is better here because it shows the full shape of the distribution (e.g., where salaries cluster, whether the distribution is skewed or bimodal), not just five summary numbers. The boxplot is simpler and better for quick side-by-side comparison of medians and outliers, but with only 10 data points per department the violin's added density detail is more informative about how salaries are actually spread within each department.

Activity 3: Histogram & Density Plot – Website Response Time

A histogram bins continuous response-time data into intervals to show frequency, while a density plot smooths this into a continuous curve estimating the underlying probability distribution.





Q1. Draw histogram.

The histogram above bins the 20 response-time observations (210–380 ms) into 8 intervals, showing that most requests fall in the 250–320 ms range.

Q2. Plot density.

The density (KDE) plot above smooths the histogram into a continuous curve, confirming a single peak (mode) roughly around 280–300 ms.

Q3. Determine skewness.

The distribution is very close to symmetric, with a skewness coefficient of about -0.008 (essentially zero). This means response times are evenly spread around the mean with no strong tail in either direction.

Q4. Common range.

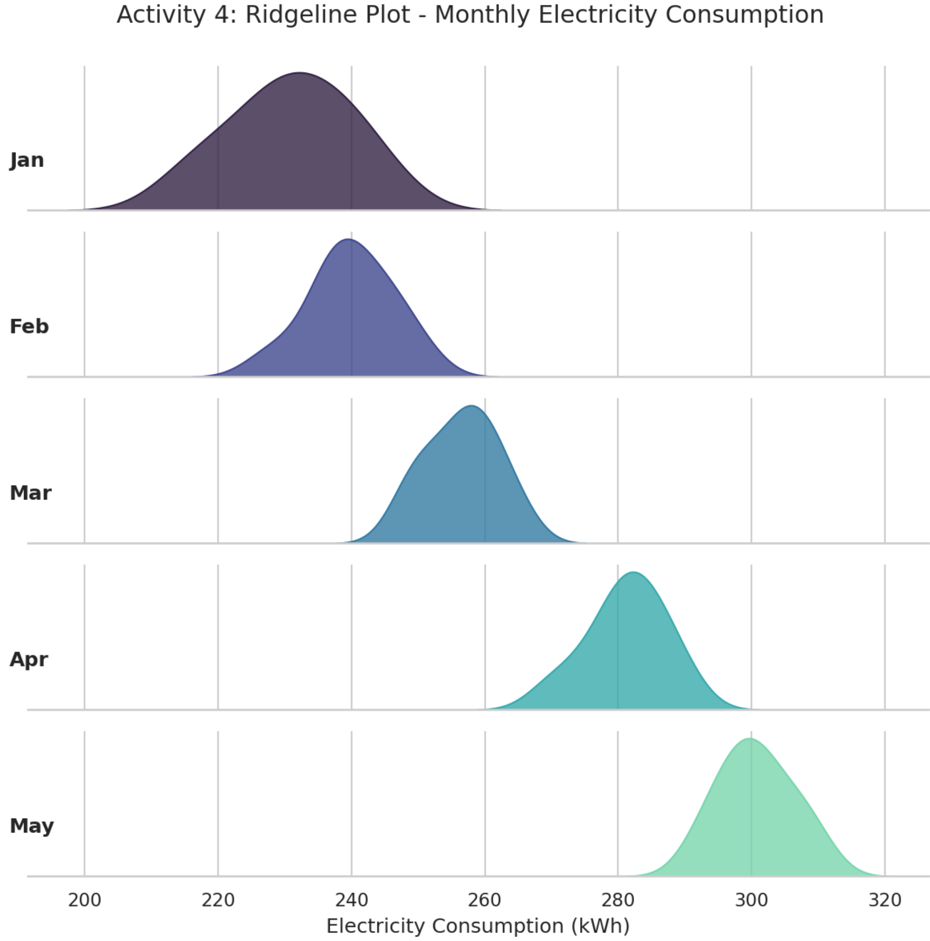
The mean response time is 292.5 ms and the median is 295 ms. Most values (roughly the middle 50%) fall between about 245 ms and 335 ms, with the bulk of observations concentrated in the 250–320 ms band.

Q5. Which plot better represents distribution?

The density plot better represents the underlying distribution shape since it smooths out the arbitrary binning choice of the histogram and gives a clearer picture of symmetry/skewness. However, the histogram is more transparent about actual counts in each range, which matters when the sample size is small (n = 20) and smoothing could hide real structure. For this dataset, the density plot is preferable for judging overall shape, but both are worth showing together.

Activity 4: Ridgeline Plot – Electricity Consumption

A ridgeline plot stacks multiple overlapping density curves (one per month here) to compare how a distribution changes across an ordered category, letting a viewer see the whole progression in a single, compact figure.



Q1. Create ridgeline plot.

The ridgeline plot above shows the density of electricity consumption across the 10 households for each month from January to May, stacked vertically in chronological order.

Q2. Highest average month?

May has the highest average consumption at 300.7 kWh, followed by April (281.2), March (256.5), February (239.9), and January (230.8).

Q3. Widest spread?

January has the widest spread (standard deviation ≈ 9.22 kWh), likely due to more variable heating/appliance usage across households at the start of the year, while later months (Feb–May) are more tightly clustered (std ≈ 5.4–6.3).

Q4. Describe trend.

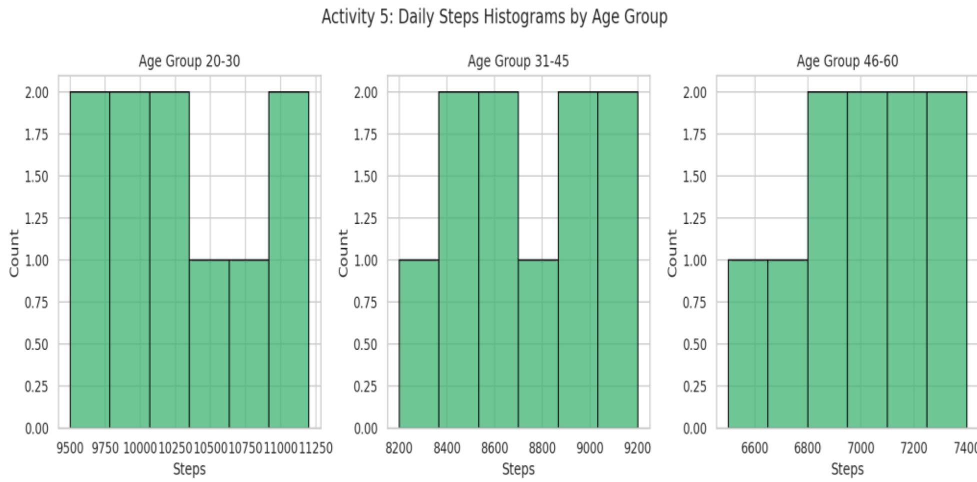
There is a clear, steady upward trend in electricity consumption from January through May across every household — consumption rises by roughly 70 kWh on average from Jan (230.8) to May (300.7), likely reflecting increasing cooling/appliance load as temperatures rise toward summer.

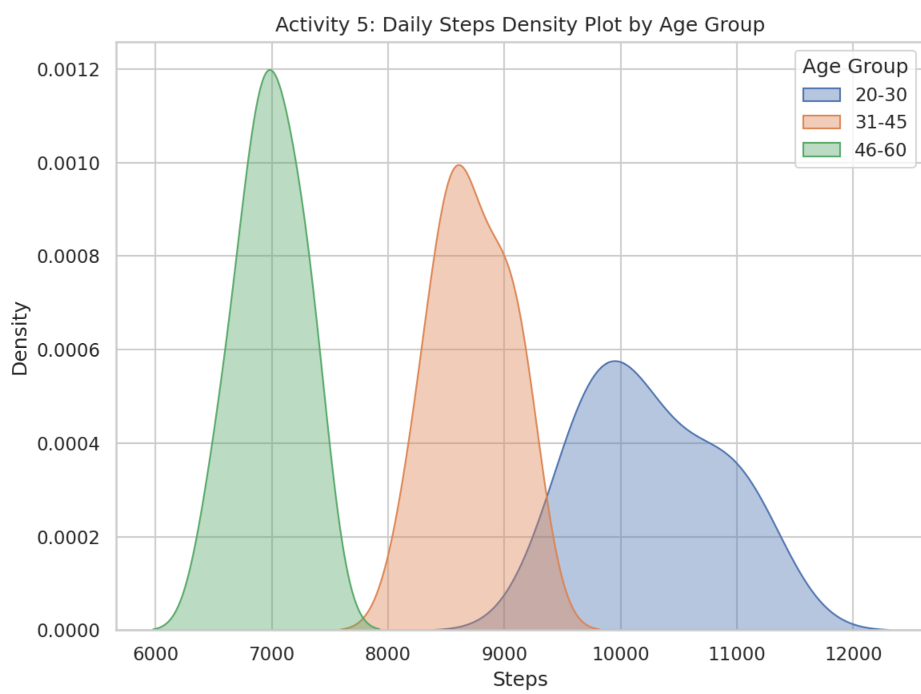
Q5. Why ridgeline?

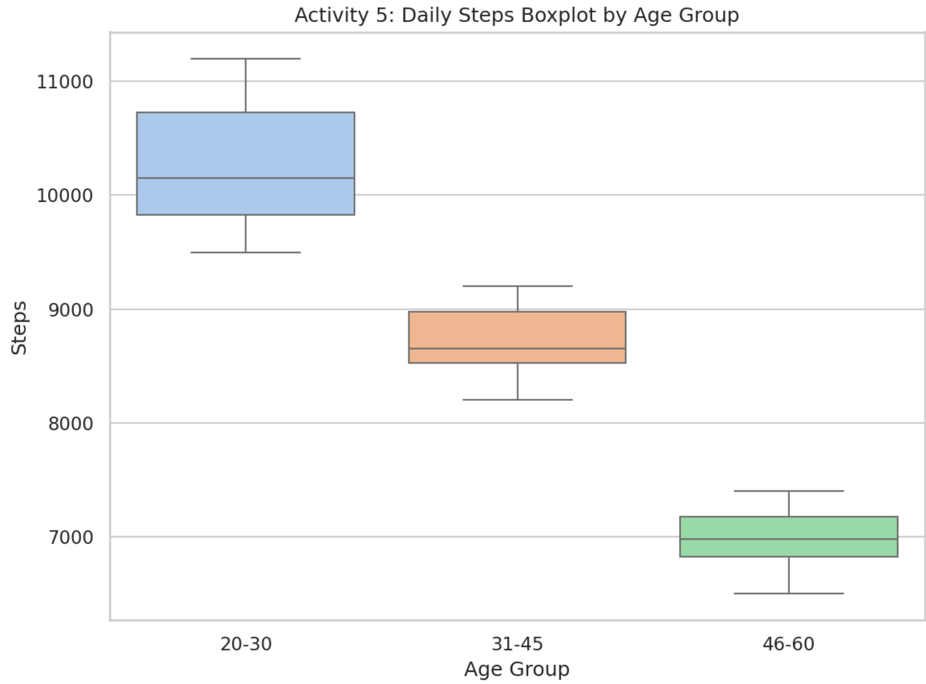
A ridgeline plot is ideal here because it lets you compare five distributions (one per month) at once while preserving their individual shapes, and the vertical stacking in time order makes the month-over-month upward shift in consumption immediately visible — something a single overlaid density plot or boxplot grid would show less clearly.

Activity 5: Histogram, Density & Boxplot – Daily Steps

This activity compares step-count distributions across three age groups using three complementary visualizations: histograms (raw frequency), density curves (smoothed shape), and boxplots (summary statistics).







Q1. Draw histograms.

The histograms above show step-count frequency for each age group: 20–30 clusters at the highest step counts (9,500–11,200), 31–45 in the middle (8,200–9,200), and 46–60 at the lowest (6,500–7,400).

Q2. Plot density curves.

The overlaid density plot confirms three clearly separated, non-overlapping distributions, showing that daily step count declines steadily as age group increases.

Q3. Create boxplots.

The boxplot above shows the median and spread for each age group side by side, making the age-related decline in step count and each group's relatively tight interquartile range visually clear.

Q4. Highest median group?

The 20–30 age group has the highest median at 10,150 steps/day, compared to 8,650 for 31–45 and 6,975 for 46–60.

Q5. Best visualization and justify.

The boxplot is the best single visualization for this dataset because the three age groups are cleanly separated and the main goal is comparing central tendency and spread across groups — a boxplot does this compactly and directly. The density plot is a strong complement since it confirms the groups don't overlap, but with only 10 observations per group, histograms are the least informative option because the small sample size makes bin counts noisy and harder to compare across the three panels.